

Volume I Foundational Geometry Reference

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Euclidean Geometry

Definitions

Definition 0.0.1 (Point). A point is that which has no part.

Definition 0.0.2 (Line). A line is breadthless length.

Definition 0.0.3 (Ends of a line). The ends of a line are points.

Definition 0.0.4 (Straight line). A straight line is a line which lies evenly with the points on itself.

Definition 0.0.5 (Surface). A surface is that which has length and breadth only.

Definition 0.0.6 (Edges of a surface). The edges of a surface are lines.

Definition 0.0.7 (Plane surface). A plane surface is a surface which lies evenly with the straight lines on itself.

Definition 0.0.8 (Plane angle). A plane angle is the inclination to one another of two lines in a plane which meet one another and do not lie in a straight line.

Definition 0.0.9 (Rectilinear angle). And when the lines containing the angle are straight, the angle is called rectilinear.

Definition 0.0.10 (Right angle and perpendicular). When a straight line standing on a straight line makes the adjacent angles equal to one another, each of the equal angles is right, and the straight line standing on the other is called a perpendicular to that on which it stands.

Definition 0.0.11 (Obtuse angle). An obtuse angle is an angle greater than a right angle.

Definition 0.0.12 (Acute angle). An acute angle is an angle less than a right angle.

Definition 0.0.13 (Boundary). A boundary is that which is an extremity of anything.

Definition 0.0.14 (Figure). A figure is that which is contained by any boundary or boundaries.

Definition 0.0.15 (Circle). A circle is a plane figure contained by one line such that all the straight lines falling upon it from one point among those lying within the figure equal one another.

Definition 0.0.16 (Center of a circle). And the point is called the center of the circle.

Definition 0.0.17 (Diameter). A diameter of the circle is any straight line drawn through the center and terminated in both directions by the circumference of the circle, and such a straight line also bisects the circle.

Definition 0.0.18 (Semicircle). A semicircle is the figure contained by the diameter and the circumference cut off by it. And the center of the semicircle is the same as that of the circle.

Definition 0.0.19 (Rectilinear figures). Rectilinear figures are those which are contained by straight lines, trilateral figures being those contained by three, quadrilateral those contained by four, and multilateral those contained by more than four straight lines.

Definition 0.0.20 (Equilateral, isosceles, and scalene triangles). Of trilateral figures, an equilateral triangle is that which has its three sides equal, an isosceles triangle that which has two of its sides alone equal, and a scalene triangle that which has its three sides unequal.

Definition 0.0.21 (Right-angled, obtuse-angled, and acute-angled triangles). Further, of trilateral figures, a right-angled triangle is that which has a right angle, an obtuse-angled triangle that which has an obtuse angle, and an acute-angled triangle that which has its three angles acute.

Definition 0.0.22 (Quadrilateral classes). Of quadrilateral figures, a square is that which is both equilateral and right-angled; an oblong that which is right-angled but not equilateral; a rhombus that which is equilateral but not right-angled; and a rhomboid that which has its opposite sides and angles equal to one another but is neither equilateral nor right-angled. And let quadrilaterals other than these be called trapezia.

Definition 0.0.23 (Parallel straight lines). Parallel straight lines are straight lines which, being in the same plane and being produced indefinitely in both directions, do not meet one another in either direction.

Postulates

0.1 Foundations

Axioms

Propositions

Theorem 0.1.1 (Euclid I.1: Constructing an equilateral triangle). *Given a finite straight line segment AB , there exists an equilateral triangle ABC constructed on AB .*

Theorem 0.1.2 (Euclid I.2: Copying a segment from a given point). *Given a point A and a finite straight line segment BC , there exists a straight line segment beginning at A and equal to BC .*

Theorem 0.1.3 (Euclid I.3: Cutting off an equal segment). *Given two unequal finite straight line segments, there exists a construction which cuts off from the greater a straight line segment equal to the less.*

Theorem 0.1.4 (Euclid I.4: Side-angle-side congruence). *If two triangles have two sides equal to two sides respectively, and have the angles contained by the equal straight lines equal, then they also have the base equal to the base, the triangle equal to the triangle, and the remaining angles equal to the remaining angles respectively.*

Theorem 0.1.5 (Euclid I.31: Drawing a parallel through a point). *Through a given point, to draw a straight line parallel to a given straight line.*

Theorem 0.1.6 (Euclid I.32: Triangle angle sum). *In any triangle, if one of the sides is produced, then the exterior angle is equal to the two interior and opposite angles, and the three interior angles of the triangle are equal to two right angles.*

Theorem 0.1.7 (Euclid I.41: Triangle and parallelogram on the same base). *If a parallelogram has the same base with a triangle and is in the same parallels, then the parallelogram is double the triangle.*

Theorem 0.1.8 (Euclid I.46: Constructing a square on a segment). *On a given straight line, to describe a square.*

Theorem 0.1.9 (Euclid I.47: Pythagorean theorem). *In right-angled triangles, the square on the side subtending the right angle is equal to the squares on the sides containing the right angle.*

0.2 Compass-and-Straightedge Constructions

Proofs

Trigonometry

0.2.1 Trigonometric Functions from Euclidean Geometry

Definition 0.2.1 (Right-triangle sine). Let T be a right triangle and let θ be one of its acute angles. Write

$$\text{opp}(T, \theta), \quad \text{adj}(T, \theta), \quad \text{hyp}(T)$$

for the side lengths opposite θ , adjacent to θ , and opposite the right angle, respectively. Define

$$\sin \theta = \frac{\text{opp}(T, \theta)}{\text{hyp}(T)}.$$

Definition 0.2.2 (Acute-angle sine). For an acute Euclidean angle θ , choose any right triangle T having θ as an acute angle and set

$$\sin \theta = \frac{\text{opp}(T, \theta)}{\text{hyp}(T)}.$$

Definition 0.2.3 (Unit-circle sine). Let θ be an angle in standard position, and let $P(\theta) = (x_\theta, y_\theta)$ be the point where its terminal ray meets the unit circle. Define

$$\sin \theta = y_\theta.$$

Definition 0.2.4 (Right-triangle cosine). Let T be a right triangle and let θ be one of its acute angles. Define

$$\cos \theta = \frac{\text{adj}(T, \theta)}{\text{hyp}(T)}.$$

Definition 0.2.5 (Acute-angle cosine). For an acute Euclidean angle θ , choose any right triangle T having θ as an acute angle and set

$$\cos \theta = \frac{\text{adj}(T, \theta)}{\text{hyp}(T)}.$$

Definition 0.2.6 (Unit-circle cosine). Let θ be an angle in standard position, and let $P(\theta) = (x_\theta, y_\theta)$ be the point where its terminal ray meets the unit circle. Define

$$\cos \theta = x_\theta.$$

Definition 0.2.7 (Right-triangle tangent). Let T be a right triangle and let θ be one of its acute angles. Define

$$\tan \theta = \frac{\text{opp}(T, \theta)}{\text{adj}(T, \theta)}.$$

Definition 0.2.8 (Extended tangent). Where $\cos \theta \neq 0$, define

$$\tan \theta = \frac{\sin \theta}{\cos \theta}.$$

Definition 0.2.9 (Right-triangle cotangent). Let T be a right triangle and let θ be one of its acute angles. Define

$$\cot \theta = \frac{\text{adj}(T, \theta)}{\text{opp}(T, \theta)}.$$

Definition 0.2.10 (Extended cotangent). Where $\sin \theta \neq 0$, define

$$\cot \theta = \frac{\cos \theta}{\sin \theta}.$$

Definition 0.2.11 (Right-triangle secant). Let T be a right triangle and let θ be one of its acute angles. Define

$$\sec \theta = \frac{\text{hyp}(T)}{\text{adj}(T, \theta)}.$$

Definition 0.2.12 (Extended secant). Where $\cos \theta \neq 0$, define

$$\sec \theta = \frac{1}{\cos \theta}.$$

Definition 0.2.13 (Right-triangle cosecant). Let T be a right triangle and let θ be one of its acute angles. Define

$$\csc \theta = \frac{\text{hyp}(T)}{\text{opp}(T, \theta)}.$$

Definition 0.2.14 (Extended cosecant). Where $\sin \theta \neq 0$, define

$$\csc \theta = \frac{1}{\sin \theta}.$$

Definition 0.2.15 (Inverse sine). The *inverse sine* function is the inverse of sine after sine is restricted to the interval

$$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right].$$

It is denoted by $\arcsin$ or $\sin^{-1}$ and has domain $[-1, 1]$ and range $[-\pi/2, \pi/2]$.

$$\arcsin x = \theta \iff \sin \theta = x \quad \text{and} \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}.$$

Theorem 0.2.16 (Principal inverse sine identities). *For every $x \in [-1, 1]$,*

$$\sin(\arcsin x) = x.$$

For every $\theta \in [-\pi/2, \pi/2]$,

$$\arcsin(\sin \theta) = \theta.$$

Definition 0.2.17 (Inverse cosine). The *inverse cosine* function is the inverse of cosine after cosine is restricted to the interval $[0, \pi]$. It is denoted by $\arccos$ or $\cos^{-1}$ and has domain $[-1, 1]$ and range $[0, \pi]$.

$$\arccos x = \theta \iff \cos \theta = x \quad \text{and} \quad 0 \leq \theta \leq \pi.$$

Theorem 0.2.18 (Principal inverse cosine identities). *For every $x \in [-1, 1]$,*

$$\cos(\arccos x) = x.$$

For every $\theta \in [0, \pi]$,

$$\arccos(\cos \theta) = \theta.$$

Definition 0.2.19 (Inverse tangent). The *inverse tangent* function is the inverse of tangent after tangent is restricted to the interval $(-\pi/2, \pi/2)$. It is denoted by $\arctan$ or $\tan^{-1}$ and has domain $\mathbb{R}$ and range $(-\pi/2, \pi/2)$.

$$\arctan x = \theta \iff \tan \theta = x \quad \text{and} \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2}.$$

Theorem 0.2.20 (Principal inverse tangent identities). *For every $x \in \mathbb{R}$,*

$$\tan(\arctan x) = x.$$

For every $\theta \in (-\pi/2, \pi/2)$,

$$\arctan(\tan \theta) = \theta.$$

Definition 0.2.21 (Inverse cotangent). The *inverse cotangent* function is the inverse of cotangent after cotangent is restricted to the interval $(0, \pi)$. It is denoted by arccot or $\cot^{-1}$ and has domain $\mathbb{R}$ and range $(0, \pi)$.

$$\operatorname{arccot} x = \theta \iff \cot \theta = x \text{ and } 0 < \theta < \pi.$$

Theorem 0.2.22 (Principal inverse cotangent identities). *For every $x \in \mathbb{R}$,*

$$\cot(\operatorname{arccot} x) = x.$$

For every $\theta \in (0, \pi)$,

$$\operatorname{arccot}(\cot \theta) = \theta.$$

Definition 0.2.23 (Inverse secant). The *inverse secant* function is the inverse of secant after secant is restricted to $[0, \pi] \setminus \{\pi/2\}$. It is denoted by arcsec or $\sec^{-1}$ and has domain $(-\infty, -1] \cup [1, \infty)$ and range $[0, \pi] \setminus \{\pi/2\}$.

$$\operatorname{arcsec} x = \theta \iff \sec \theta = x \text{ and } 0 \leq \theta \leq \pi, \theta \neq \frac{\pi}{2}.$$

Theorem 0.2.24 (Principal inverse secant identities). *For every $x \in (-\infty, -1] \cup [1, \infty)$,*

$$\sec(\operatorname{arcsec} x) = x.$$

For every $\theta \in [0, \pi] \setminus \{\pi/2\}$,

$$\operatorname{arcsec}(\sec \theta) = \theta.$$

Definition 0.2.25 (Inverse cosecant). The *inverse cosecant* function is the inverse of cosecant after cosecant is restricted to $[-\pi/2, \pi/2] \setminus \{0\}$. It is denoted by arccsc or $\csc^{-1}$ and has domain $(-\infty, -1] \cup [1, \infty)$ and range $[-\pi/2, \pi/2] \setminus \{0\}$.

$$\operatorname{arccsc} x = \theta \iff \csc \theta = x \text{ and } -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}, \theta \neq 0.$$

Theorem 0.2.26 (Principal inverse cosecant identities). *For every $x \in (-\infty, -1] \cup [1, \infty)$,*

$$\csc(\operatorname{arccsc} x) = x.$$

For every $\theta \in [-\pi/2, \pi/2] \setminus \{0\}$,

$$\operatorname{arccsc}(\csc \theta) = \theta.$$

Theorem 0.2.27 (Cofunction identities for inverse sine and inverse cosine). *For every $x \in [-1, 1]$,*

$$\arcsin x + \arccos x = \frac{\pi}{2}.$$

Equivalently,

$$\arccos x = \frac{\pi}{2} - \arcsin x, \quad \arcsin x = \frac{\pi}{2} - \arccos x.$$

Theorem 0.2.28 (Reciprocal identities for inverse secant and inverse cosecant). *For every $x \in (-\infty, -1] \cup [1, \infty)$,*

$$\operatorname{arcsec} x = \arccos\left(\frac{1}{x}\right), \quad \operatorname{arccsc} x = \arcsin\left(\frac{1}{x}\right).$$

Theorem 0.2.29 (Reciprocal identity for inverse cotangent). *For every nonzero real number x ,*

$$\operatorname{arccot} x = \begin{cases} \arctan(1/x), & x > 0, \\ \pi + \arctan(1/x), & x < 0. \end{cases}$$

Theorem 0.2.30 (Similarity invariance of trigonometric ratios). *If two right triangles have the same acute angle θ , then their corresponding opposite-to-hypotenuse and adjacent-to-hypotenuse ratios are equal.*

Theorem 0.2.31 (Well-definedness of sine and cosine). *For each acute Euclidean angle θ , the values $\sin \theta$ and $\cos \theta$ are independent of the right triangle chosen to compute them.*

Theorem 0.2.32 (Six trigonometric functions). *The six right-triangle trigonometric ratios define functions of an acute Euclidean angle, subject to the denominator restrictions in their defining quotients.*

Theorem 0.2.33 (Unit-circle agreement with right-triangle trigonometry). *For $0 < \theta < 90^\circ$, the unit-circle definitions of sine and cosine agree with the right-triangle definitions.*

Definition 0.2.34 (Radian measure). One radian is the angle that cuts off arc length 1 on the unit circle. A full turn has measure 2π radians.

Theorem 0.2.35 (Symmetry and periodicity of sine and cosine). *For every angle in the unit-circle domain,*

$$\cos(-\theta) = \cos \theta, \quad \sin(-\theta) = -\sin \theta,$$

and one full turn returns the same sine and cosine values.

Theorem 0.2.36 (Pythagorean identity). *For every angle in the unit-circle domain,*

$$\cos^2 \theta + \sin^2 \theta = 1.$$

Theorem 0.2.37 (Reciprocal identities). *Where both sides are defined,*

$$\sec \theta = \frac{1}{\cos \theta}, \quad \csc \theta = \frac{1}{\sin \theta}, \quad \cot \theta = \frac{1}{\tan \theta}.$$

Theorem 0.2.38 (Quotient identities). *Where both sides are defined,*

$$\tan \theta = \frac{\sin \theta}{\cos \theta}, \quad \cot \theta = \frac{\cos \theta}{\sin \theta}.$$

Theorem 0.2.39 (Tangent and secant Pythagorean identities). *Where both sides are defined,*

$$1 + \tan^2 \theta = \sec^2 \theta, \quad 1 + \cot^2 \theta = \csc^2 \theta.$$

Theorem 0.2.40 (Angle-sum identities). *For angles in the unit-circle domain,*

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta,$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta.$$

Theorem 0.2.41 (Angle-difference identities). *For angles in the unit-circle domain,*

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta,$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta.$$

Theorem 0.2.42 (Double-angle identities). *For angles in the unit-circle domain,*

$$\sin(2\theta) = 2 \sin \theta \cos \theta,$$

$$\cos(2\theta) = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta.$$

Theorem 0.2.43 (Half-angle identities). *Where the signs are chosen according to the quadrant of $\theta/2$,*

$$\sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}}, \quad \cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}}.$$

Theorem 0.2.44 (Product-to-sum identities). *For angles in the unit-circle domain,*

$$\begin{aligned}\sin \alpha \cos \beta &= \frac{\sin(\alpha + \beta) + \sin(\alpha - \beta)}{2}, \\ \cos \alpha \cos \beta &= \frac{\cos(\alpha + \beta) + \cos(\alpha - \beta)}{2}, \\ \sin \alpha \sin \beta &= \frac{\cos(\alpha - \beta) - \cos(\alpha + \beta)}{2}.\end{aligned}$$

Theorem 0.2.45 (Sum-to-product identities). *For angles in the unit-circle domain,*

$$\begin{aligned}\sin u + \sin v &= 2 \sin \frac{u+v}{2} \cos \frac{u-v}{2}, \\ \cos u + \cos v &= 2 \cos \frac{u+v}{2} \cos \frac{u-v}{2}, \\ \cos u - \cos v &= -2 \sin \frac{u+v}{2} \sin \frac{u-v}{2}.\end{aligned}$$

0.2.2 Hyperbolic Trigonometric Functions from Geometry

Definition 0.2.46 (Hyperbolic sector). Let $O = (0, 0)$ and $V = (1, 0)$. For a point $P = (u, v)$ on the right branch of $x^2 - y^2 = 1$, write

$$\text{HArea}(O, V, P)$$

for the signed area of the region bounded by the segment OV , the segment OP , and the hyperbolic arc from V to P . The sign is positive on the upper branch and negative on the lower branch.

Definition 0.2.47 (Swept hyperbola point). For $t \geq 0$, let P_t be the unique point on the upper right branch of $x^2 - y^2 = 1$ satisfying

$$2 \text{HArea}(O, V, P_t) = t.$$

For $t < 0$, let P_t be the reflection of P_{-t} across the x -axis.

Definition 0.2.48 (Hyperbolic sine). For each $t \in \mathbb{R}$, define $\sinh t$ to be the second coordinate of the swept hyperbola point P_t :

$$P_t = (u, \sinh t).$$

Definition 0.2.49 (Hyperbolic cosine). For each $t \in \mathbb{R}$, define $\cosh t$ to be the first coordinate of the swept hyperbola point P_t :

$$P_t = (\cosh t, v).$$

Definition 0.2.50 (Hyperbolic tangent). For each $t \in \mathbb{R}$, define the *hyperbolic tangent* by

$$\tanh t = \frac{\sinh t}{\cosh t}.$$

Definition 0.2.51 (Hyperbolic cotangent). For each $t \in \mathbb{R} \setminus \{0\}$, define the *hyperbolic cotangent* by

$$\coth t = \frac{\cosh t}{\sinh t}.$$

Definition 0.2.52 (Hyperbolic secant). For each $t \in \mathbb{R}$, define the *hyperbolic secant* by

$$\operatorname{sech} t = \frac{1}{\cosh t}.$$

Definition 0.2.53 (Hyperbolic cosecant). For each $t \in \mathbb{R} \setminus \{0\}$, define the *hyperbolic cosecant* by

$$\operatorname{csch} t = \frac{1}{\sinh t}.$$

Definition 0.2.54 (Inverse hyperbolic sine). The *inverse hyperbolic sine* function is the inverse of $\sinh$ on its full domain. It is denoted by arsinh and has domain $\mathbb{R}$ and range $\mathbb{R}$.

$$\operatorname{arsinh} x = t \iff \sinh t = x.$$

Theorem 0.2.55 (Principal inverse hyperbolic sine identities). *For every $x \in \mathbb{R}$,*

$$\sinh(\operatorname{arsinh} x) = x.$$

For every $t \in \mathbb{R}$,

$$\operatorname{arsinh}(\sinh t) = t.$$

Definition 0.2.56 (Inverse hyperbolic cosine). The *inverse hyperbolic cosine* function is the inverse of $\cosh$ after $\cosh$ is restricted to the interval $[0, \infty)$. It is denoted by arcosh and has domain $[1, \infty)$ and range $[0, \infty)$.

$$\operatorname{arcosh} x = t \iff \cosh t = x \text{ and } t \geq 0.$$

Theorem 0.2.57 (Principal inverse hyperbolic cosine identities). *For every $x \in [1, \infty)$,*

$$\cosh(\operatorname{arcosh} x) = x.$$

For every $t \in [0, \infty)$,

$$\operatorname{arcosh}(\cosh t) = t.$$

Definition 0.2.58 (Inverse hyperbolic tangent). The *inverse hyperbolic tangent* function is the inverse of $\tanh$ on its full domain. It is denoted by artanh and has domain $(-1, 1)$ and range $\mathbb{R}$.

$$\operatorname{artanh} x = t \iff \tanh t = x.$$

Theorem 0.2.59 (Principal inverse hyperbolic tangent identities). *For every $x \in (-1, 1)$,*

$$\tanh(\operatorname{artanh} x) = x.$$

For every $t \in \mathbb{R}$,

$$\operatorname{artanh}(\tanh t) = t.$$

Definition 0.2.60 (Inverse hyperbolic cotangent). The *inverse hyperbolic cotangent* function is the inverse of $\coth$ on its two natural branches. It is denoted by arcoth and has domain $(-\infty, -1) \cup (1, \infty)$ and range $\mathbb{R} \setminus \{0\}$.

$$\operatorname{arcoth} x = t \iff \coth t = x \text{ and } t \neq 0.$$

Theorem 0.2.61 (Principal inverse hyperbolic cotangent identities). *For every $x \in (-\infty, -1) \cup (1, \infty)$,*

$$\coth(\operatorname{arcoth} x) = x.$$

For every $t \in \mathbb{R} \setminus \{0\}$,

$$\operatorname{arcoth}(\coth t) = t.$$

Definition 0.2.62 (Inverse hyperbolic secant). The *inverse hyperbolic secant* function is the inverse of sech after sech is restricted to $[0, \infty)$. It is denoted by arsech and has domain $(0, 1]$ and range $[0, \infty)$.

$$\operatorname{arsech} x = t \iff \operatorname{sech} t = x \text{ and } t \geq 0.$$

Theorem 0.2.63 (Principal inverse hyperbolic secant identities). *For every $x \in (0, 1]$,*

$$\operatorname{sech}(\operatorname{arsech} x) = x.$$

For every $t \in [0, \infty)$,

$$\operatorname{arsech}(\operatorname{sech} t) = t.$$

Definition 0.2.64 (Inverse hyperbolic cosecant). The *inverse hyperbolic cosecant* function is the inverse of csch on its full domain. It is denoted by arcsch and has domain $\mathbb{R} \setminus \{0\}$ and range $\mathbb{R} \setminus \{0\}$.

$$\operatorname{arcsch} x = t \iff \operatorname{csch} t = x \text{ and } t \neq 0.$$

Theorem 0.2.65 (Principal inverse hyperbolic cosecant identities). *For every $x \in \mathbb{R} \setminus \{0\}$,*

$$\operatorname{csch}(\operatorname{arcsch} x) = x.$$

For every $t \in \mathbb{R} \setminus \{0\}$,

$$\operatorname{arcsch}(\operatorname{csch} t) = t.$$

Theorem 0.2.66 (Existence of hyperbolic sector points). *For every $t \geq 0$, there exists a point $P = (u, v)$ on the upper right branch of $x^2 - y^2 = 1$ with*

$$2 \operatorname{HArea}(O, V, P) = t.$$

Theorem 0.2.67 (Uniqueness of hyperbolic sector points). *For every $t \geq 0$, the point P in the [existence theorem](#) is unique.*

Corollary 0.2.68 (Well-definedness of hyperbolic sine and cosine). *The assignments $\cosh, \sinh : \mathbb{R} \rightarrow \mathbb{R}$ in the [definition of hyperbolic cosine](#) and the [definition of hyperbolic sine](#) are well-defined functions.*

Theorem 0.2.69 (Fundamental hyperbolic identity). *For every $t \in \mathbb{R}$,*

$$\cosh^2 t - \sinh^2 t = 1.$$

Theorem 0.2.70 (Hyperbolic Pythagorean identity). *For every $t \in \mathbb{R}$,*

$$\cosh^2 t - \sinh^2 t = 1.$$

Theorem 0.2.71 (Exponential form of hyperbolic sine and cosine). *For every $t \in \mathbb{R}$,*

$$\cosh t = \frac{e^t + e^{-t}}{2}, \quad \sinh t = \frac{e^t - e^{-t}}{2}.$$

Theorem 0.2.72 (Hyperbolic sum formulas). *For all $a, b \in \mathbb{R}$,*

$$\cosh(a + b) = \cosh a \cosh b + \sinh a \sinh b,$$

$$\sinh(a + b) = \sinh a \cosh b + \cosh a \sinh b.$$

Theorem 0.2.73 (Hyperbolic double-angle formulas). *For every $t \in \mathbb{R}$,*

$$\sinh(2t) = 2 \sinh t \cosh t,$$

$$\cosh(2t) = \cosh^2 t + \sinh^2 t = 2 \cosh^2 t - 1 = 1 + 2 \sinh^2 t.$$

Theorem 0.2.74 (Hyperbolic tangent sum formula). *For all $a, b \in \mathbb{R}$,*

$$\tanh(a + b) = \frac{\tanh a + \tanh b}{1 + \tanh a \tanh b}.$$

Theorem 0.2.75 (Logarithmic form of inverse hyperbolic sine). *For every $x \in \mathbb{R}$,*

$$\operatorname{arsinh} x = \log\left(x + \sqrt{x^2 + 1}\right).$$

Theorem 0.2.76 (Logarithmic form of inverse hyperbolic cosine). *For every $x \in [1, \infty)$,*

$$\operatorname{arcosh} x = \log\left(x + \sqrt{x^2 - 1}\right).$$

Theorem 0.2.77 (Logarithmic form of inverse hyperbolic tangent). *For every $x \in (-1, 1)$,*

$$\operatorname{artanh} x = \frac{1}{2} \log \left(\frac{1+x}{1-x} \right).$$

Theorem 0.2.78 (Logarithmic form of inverse hyperbolic cotangent). *For every $x \in (-\infty, -1) \cup (1, \infty)$,*

$$\operatorname{arcoth} x = \frac{1}{2} \log \left(\frac{x+1}{x-1} \right).$$

Theorem 0.2.79 (Reciprocal form of inverse hyperbolic secant). *For every $x \in (0, 1]$,*

$$\operatorname{arsech} x = \operatorname{arcosh} \left(\frac{1}{x} \right).$$

Theorem 0.2.80 (Reciprocal form of inverse hyperbolic cosecant). *For every $x \in \mathbb{R} \setminus \{0\}$,*

$$\operatorname{arcsch} x = \operatorname{arsinh} \left(\frac{1}{x} \right).$$

Theorem 0.2.81 (Continuity of hyperbolic sine and cosine). *The functions $\sinh$ and $\cosh$ are continuous on $\mathbb{R}$.*

Theorem 0.2.82 (Continuity of the derived hyperbolic functions). *The functions $\tanh$ and sech are continuous on $\mathbb{R}$, and $\coth$ and csch are continuous on $\mathbb{R} \setminus \{0\}$.*

Theorem 0.2.83 (Derivatives of hyperbolic sine and cosine). *For every $t \in \mathbb{R}$,*

$$\frac{d}{dt} \sinh t = \cosh t, \quad \frac{d}{dt} \cosh t = \sinh t.$$

Theorem 0.2.84 (Derivative of hyperbolic tangent). *For every $t \in \mathbb{R}$,*

$$\frac{d}{dt} \tanh t = \operatorname{sech}^2 t = 1 - \tanh^2 t.$$

Theorem 0.2.85 (Derivative of hyperbolic cotangent). *For every $t \neq 0$,*

$$\frac{d}{dt} \coth t = -\operatorname{csch}^2 t = 1 - \coth^2 t.$$

Theorem 0.2.86 (Derivative of hyperbolic secant). *For every $t \in \mathbb{R}$,*

$$\frac{d}{dt} \operatorname{sech} t = -\operatorname{sech} t \tanh t.$$

Theorem 0.2.87 (Derivative of hyperbolic cosecant). *For every $t \neq 0$,*

$$\frac{d}{dt} \operatorname{csch} t = -\operatorname{csch} t \coth t.$$

Proofs

Analytical Geometry

0.3 Analytical Geometry

0.4 Lines

Definition 0.4.1 (Point-slope form). Let $P_1 = (x_1, y_1)$ be a point in the Euclidean plane and let $m \in \mathbb{R}$. The point-slope form of the line through P_1 with slope m is

$$y - y_1 = m(x - x_1).$$

Definition 0.4.2 (Two-point form). Let $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ be distinct points with $x_1 \neq x_2$. The two-point form of the line through P_1 and P_2 is

$$y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1} \right) (x - x_1).$$

Definition 0.4.3 (General linear form). The general linear form of a line in $\mathbb{R}^2$ is

$$Ax + By + C = 0,$$

where $A, B, C \in \mathbb{R}$ and $A^2 + B^2 \neq 0$.

Definition 0.4.4 (Vector parametric form). A line in $\mathbb{R}^2$ may be represented as the image of $\mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}^2$ given by

$$\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v},$$

where $\mathbf{r}_0 \in \mathbb{R}^2$ is a point on the line and $\mathbf{v} \in \mathbb{R}^2$ is a nonzero direction vector.

Definition 0.4.5 (Intercept form). If a line meets the x -axis at $(a, 0)$ and the y -axis at $(0, b)$ with $a, b \neq 0$, its intercept form is

$$\frac{x}{a} + \frac{y}{b} = 1.$$

Definition 0.4.6 (Affine parametric line in $\mathbb{R}^n$). Given distinct points $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^n$, the line through them is

$$\mathbf{x}(t) = \mathbf{x}_1 + t(\mathbf{x}_2 - \mathbf{x}_1), \quad t \in \mathbb{R}.$$

Definition 0.4.7 (Normal form). A line in $\mathbb{R}^2$ may be written in normal form as

$$x \cos \alpha + y \sin \alpha - p = 0,$$

where $p \geq 0$ is the distance from the origin to the line.

Definition 0.4.8 (Symmetric form). Given a point (x_1, y_1) and direction cosines $(\cos \theta, \sin \theta)$, the symmetric form of a line is

$$\frac{x - x_1}{\cos \theta} = \frac{y - y_1}{\sin \theta} = r.$$

Proofs